

CHRONIC RHINOSINUSITIS

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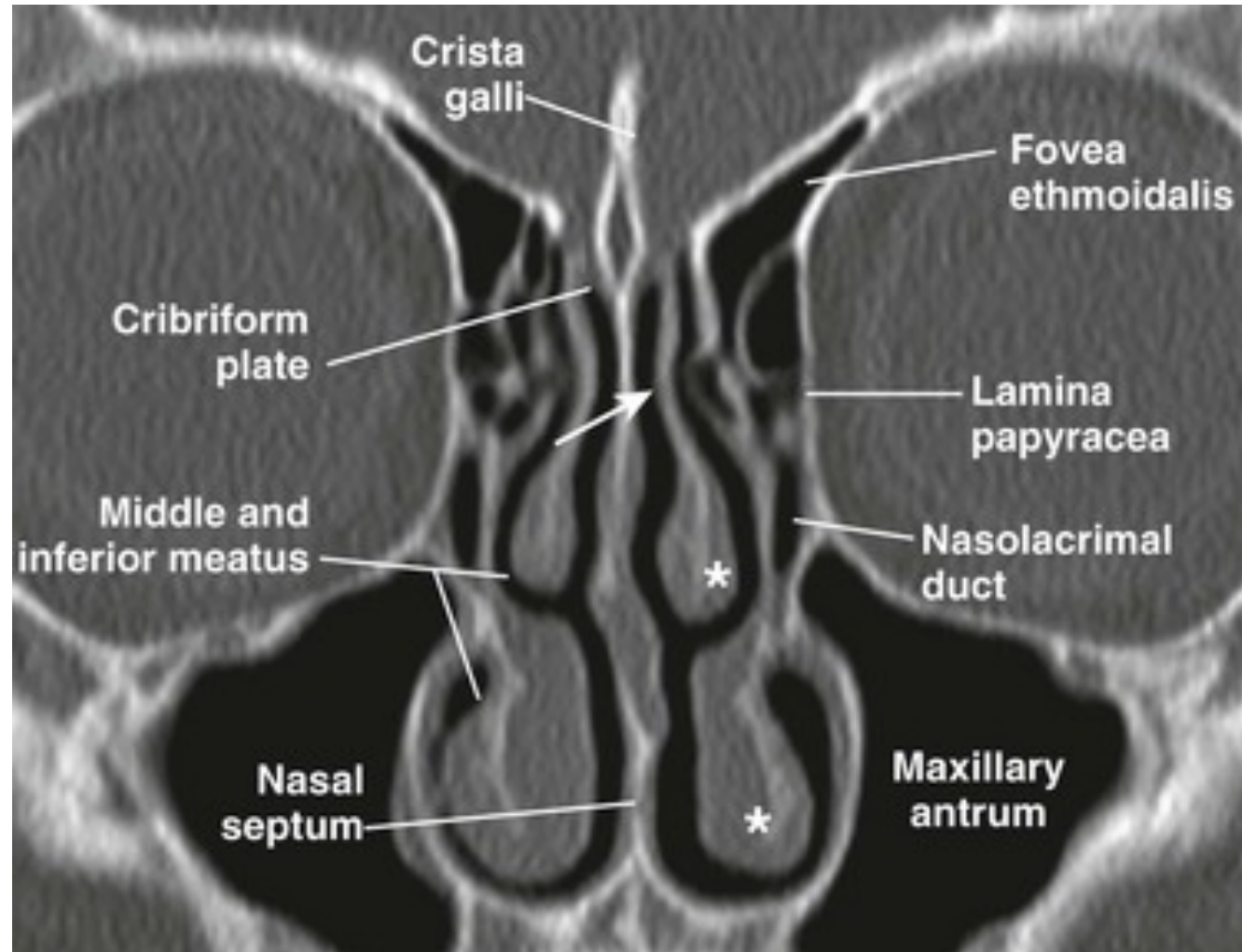
Branch of clinical science.

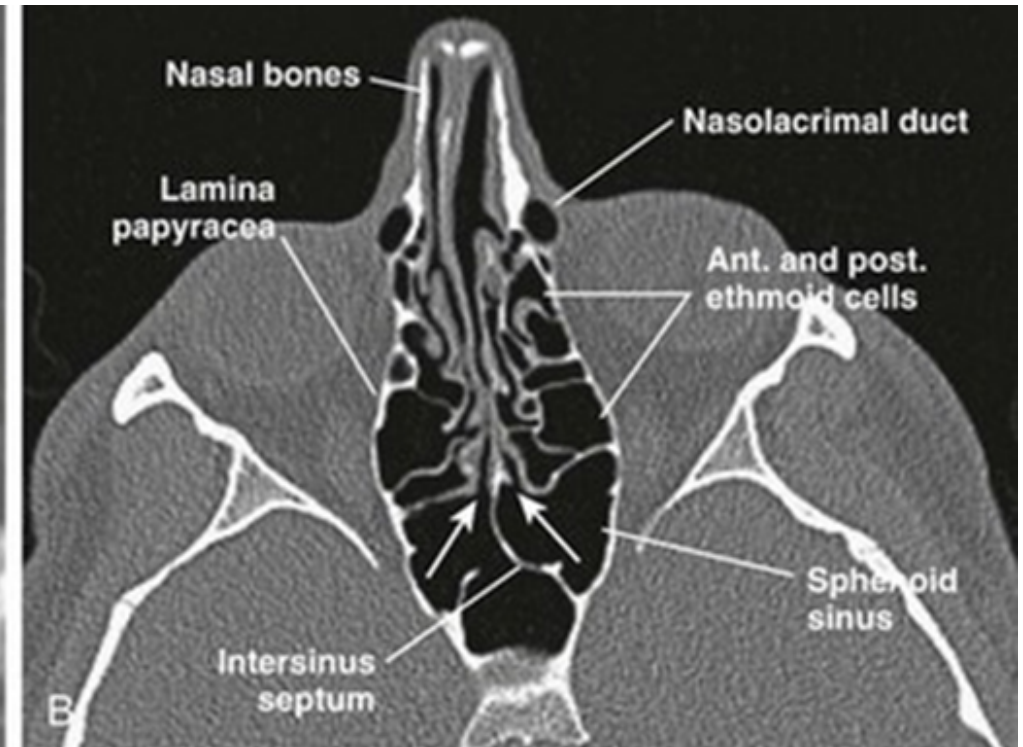
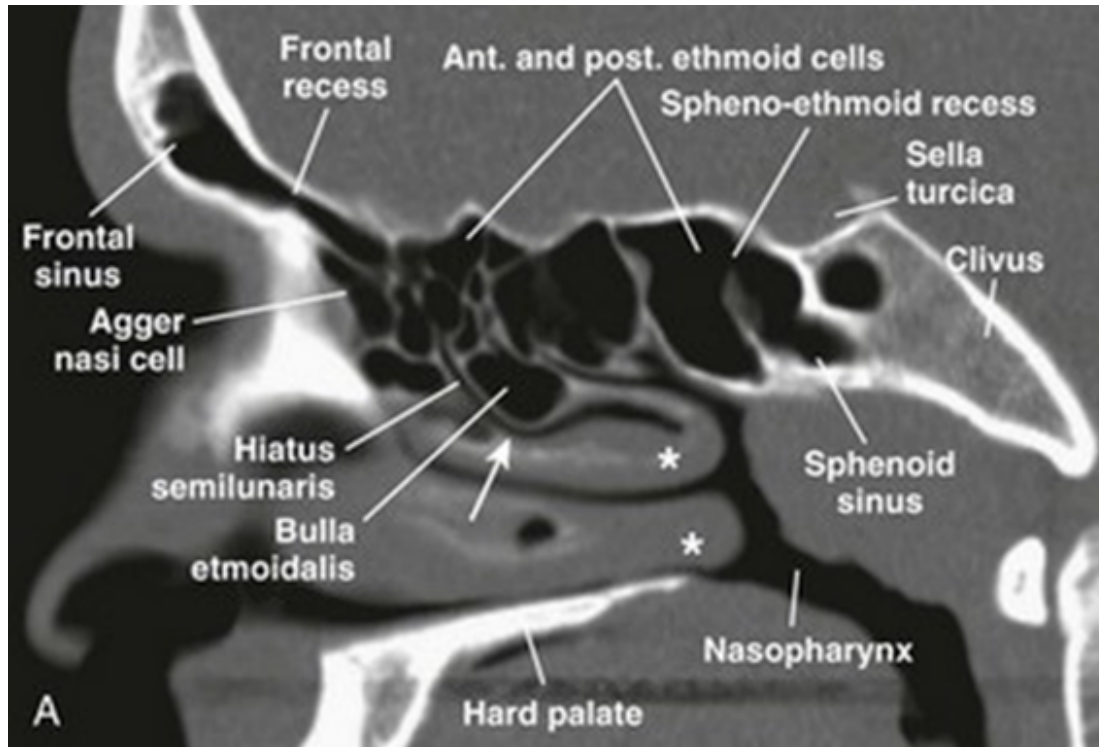
Learning outcome

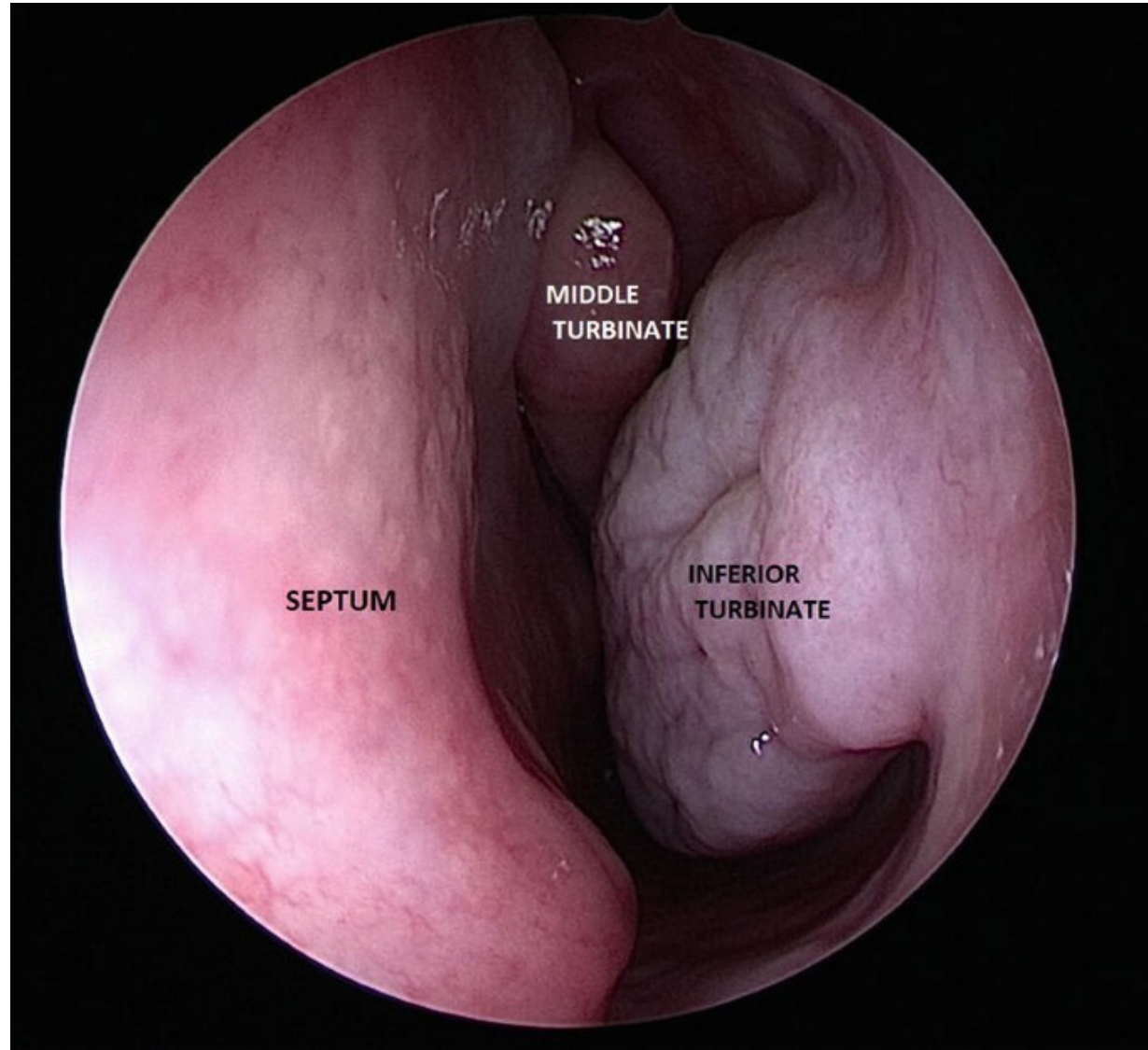
- Define Chronic Rhinosinusitis (CRS) and identify its diagnostic criteria.
- Differentiate between CRSsNP (without polyps) and CRSwNP (with polyps) subtypes.
- Recognize the causative factors and pathophysiology of CRS.
- Discuss the medical and surgical management options for CRS, including FESS.
- Understand the complications and principles of endoscopic sinus surgery.

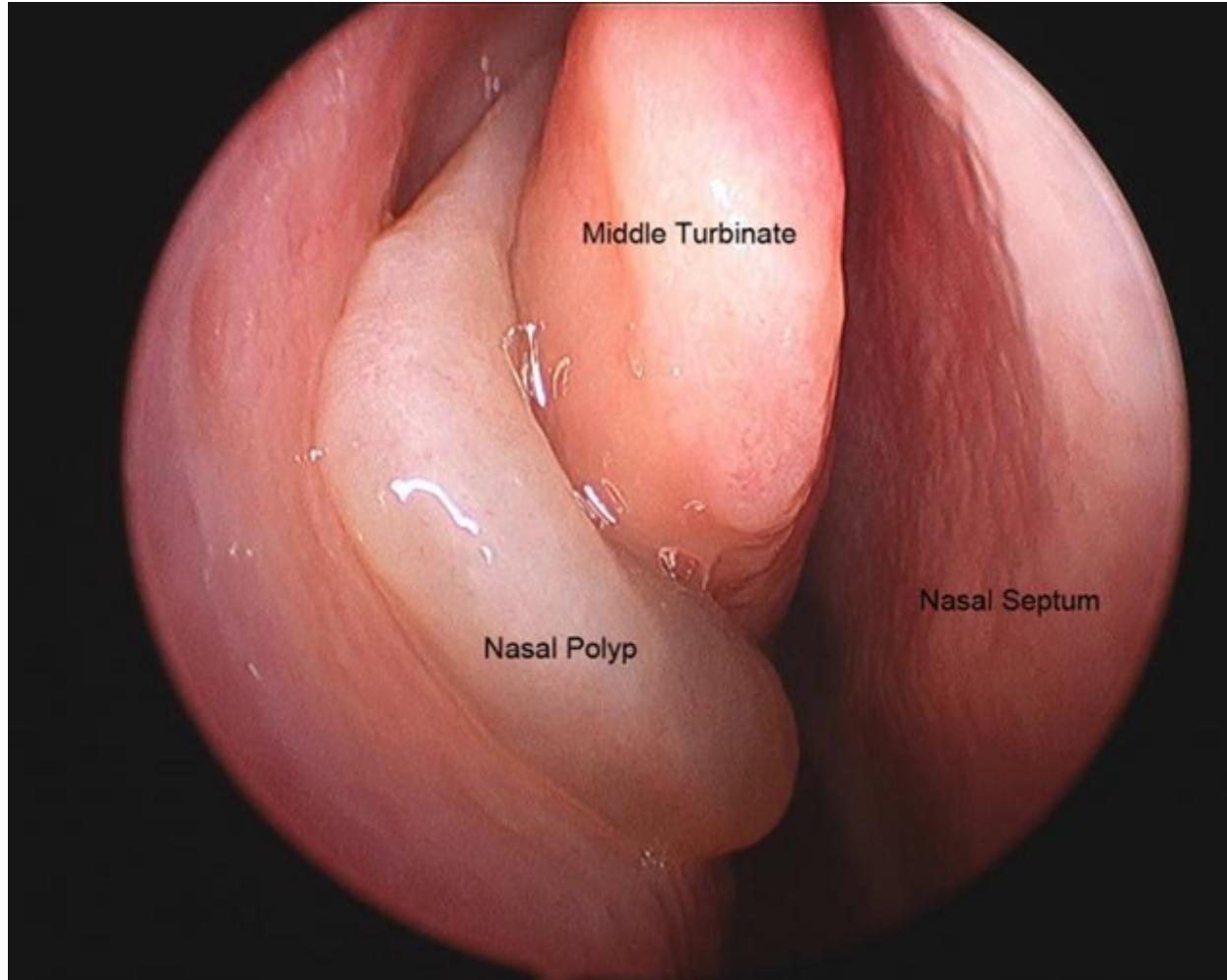
Chronic Rhinosinusitis

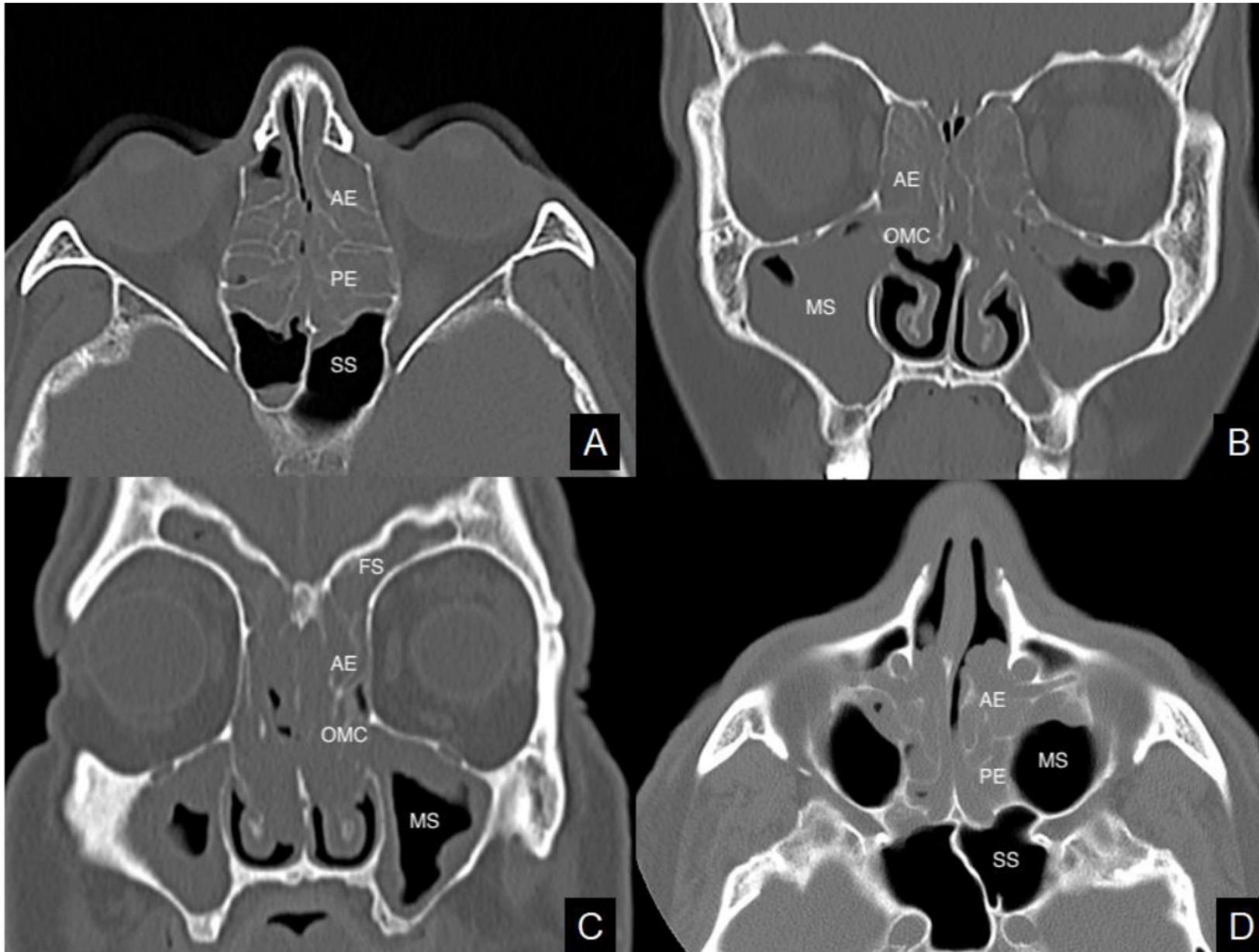
- inflammation of the nose and paranasal sinuses for $>$ or 12 weeks associated with two or more symptoms.
- one of which should be nasal blockage/obstruction/congestion or nasal discharge:
- $+/-$ facial pain/pressure , reduction or loss of smell and either
- endoscopic evidence of polyps and/or mucopurulent discharge from the middle meatus or edema, mucosal obstruction primarily in the middle meatus and/or
- CT changes: mucosal changes within the osteomeatal complex and/or sinuses











CHRONIC RHINOSINUSITIS

- It usually follows acute sinusitis that has not been adequately treated, or it may also follow a cold or tooth infection.
- It occurs when the self-cleansing mechanism of the nose and paranasal sinuses gets impaired.
- The maxillary sinus is most commonly involved.
- Organisms isolated in CRS include *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, and *Escherichia coli*. Anaerobic organisms are also found.

Classification(Sub-types of CRS)

CRSsNP-without polyp

- Facial pain, pressure, and/or fullness

Organism : S.pneumoniae, H.influenzae,
M.catarrhalis, S.aureus, S.coagulase-negative

- Glandular hyperplasia and submucosal fibrosis

CRSwNP-with polyp

- Hyposmia/anosmia

- Nasal polyps are typically bilateral
associated with AERD

- The predominance of eosinophils, high levels of histamine,
and Th2 cytokines

AFRS(allergic fungal rhinosinusitis)

- Presence of allergic mucin (thick mucus, light tan to brown to dark green, degranulated eosinophil)
- fungal hyphae in mucin
- evidence of IgE-mediated fungal allergy

Sinus surgery is usually required

Usually have nasal polyps and immunocompetent patient

Pathophysiology:chronic, allergic inflammation directed against colonizing fungi

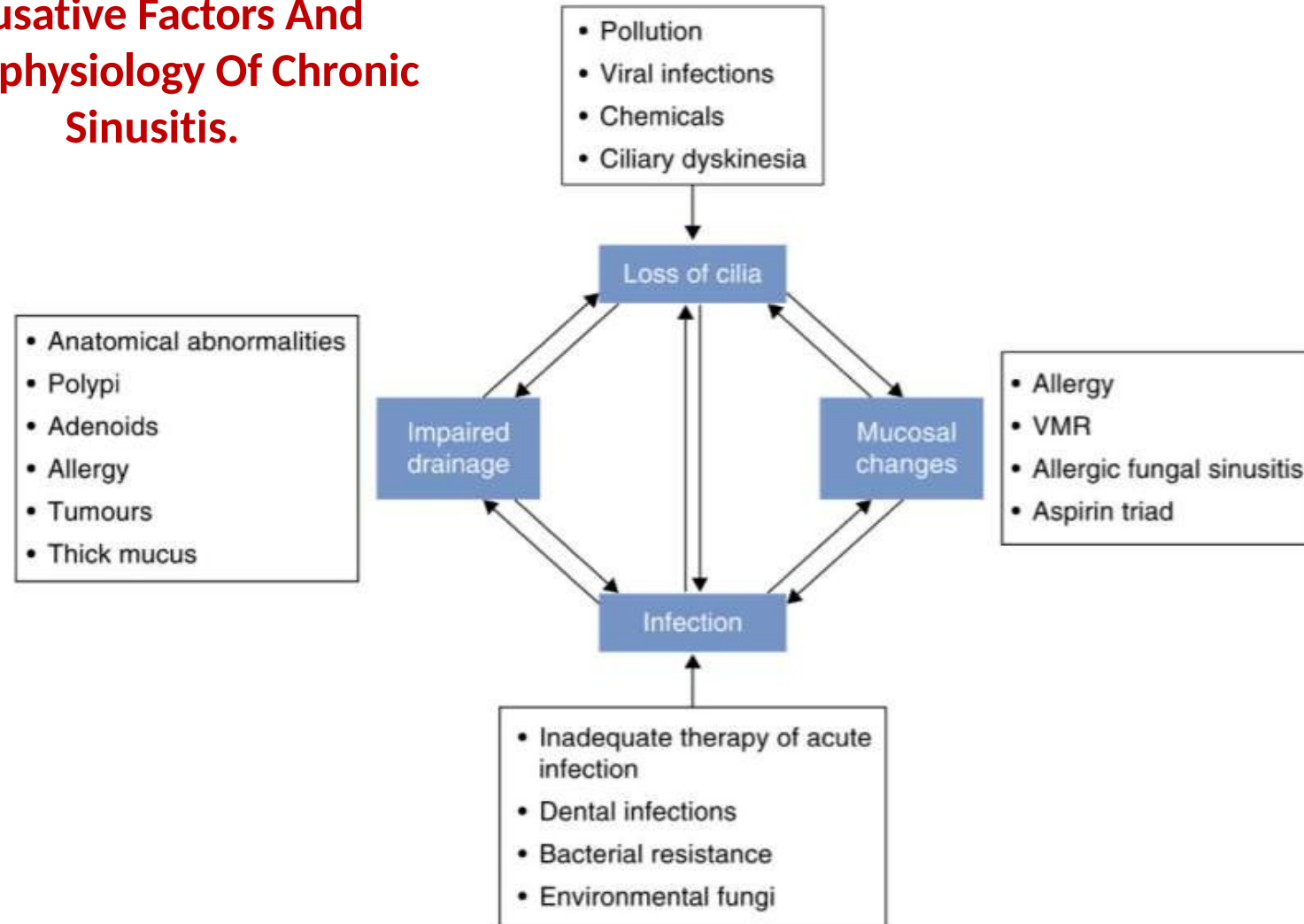
FACTORS ASSOCIATED WITH CHRONIC RHINOSINUSITIS

Systemic host factors	Local host factors	Environmental factors
1. Allergy 2. Immunodeficiency 3. Mucociliary dysfunction 4. Cystic fibrosis 5. Granulomatous diseases 6. GERD 7. Aspirin intolerance	1. Anatomic 2. Neoplasm 3. Acquired mucociliary dysfunction 4. Previous trauma or surgery	1. Microorganisms (bacteria, fungi, virus) 2. Noxious chemicals 3. Medications

Figure 2 Factors associated with CRS.

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Causative Factors And Pathophysiology Of Chronic Sinusitis.



A-Medical treatment:-

1-corticosteroid:

- topical nasal steroid sprays like fluticasone, and mometasone, are recommended for all forms of CRS
- systemic steroid: prednisolone tablet 0.5-1mg/kg in tapering dose
 - for refractory cases or unfit for surgery
 - before surgery to reduce polyp size

2-Antihistamines: helpful in allergic rhinitis

3-Antibiotics: used to treat the infection if nasal purulence is present (acute exacerbation) or to modulate immunity like macrolides and doxycycline

4-Antifungals: Indicated only in invasive forms of sinus mycosis

5-Nasal saline irrigation:

- it improves symptom scores and control

6-Mucus modifiers

- theoretically improving mucociliary transport
- Guaifenesin 1200 mg twice daily reduced congestion, postnasal drainage

7-Anticholinergics

- blocks parasympathetic input to mucus glands and reduces rhinorrhea
- may lead to the increased thickness of secretions and paradoxically worsen postnasal drainage

- 8-leukotrienes inhibitors:
 - Leukotriene receptor antagonist (montelukast) can improve symptom scores in CRS patients.
 - 5-Lipoxygenase inhibitor (zileuton) significant improvement in olfaction in patients with CRS and concomitant aspirin-sensitive asthma
- 9-Decongestants (little role in CRS)
- Topical
- Systemic
- 10-Lifestyle modification
 - Stop smoking, get adequate sleep, exercise regularly, avoid pollution, plenty of fluid

B-Surgery :

- Conventional surgery especially for nasal polyp, septoplasty
- Functional endoscopic sinus surgery(FESS):procedure of choice for refractory CRS.

Indications

- Anatomic blockage—deviated septum, enlarged turbinate, concha bullosa
- Mucocele
- Orbital abscess
- Fungal sinusitis—allergic vs. invasive (mucor)
- Tumor of nasal cavity or sinus
- Chronic, recurrent sinusitis
- Failure to respond to maximal medical therapy
- Obtain cultures

Principal goal of FESS

- Restore patency to the ostiomeatal unit
- Correction of septal deformities
- Removal of severe concha bullosa deformity
- Restoration of patency to the frontal sinus
- Delivery of the drug into the sinus
- Restoring mucocilliary clearance

Complication of FESS

Minor

- Orbital-
 - orbital emphysema orbital ecchymosis
- Nasolacrimal Duct Injury(epiphora)
- Disturbance in olfaction
- Dental pain/lip pain or numbness

Major

- Orbital hematoma -Optic nerve injury
- CSF leak
- brain laceration
- hemorrhage
 - Ethmoid artery injury
 - internal carotid artery to cavernous sinus fistula
 - sphenopalatine artery injury

Conclusion

- Chronic Rhinosinusitis is a persistent inflammatory condition of the nose and paranasal sinuses lasting more than 12 weeks.
- It often results from impaired mucociliary clearance and may follow infection or allergic inflammation.
- Accurate diagnosis, based on symptoms, endoscopic findings, and imaging, is essential for effective treatment.
- Medical therapy remains first-line, but surgery (FESS) is vital for refractory cases.
- Successful management requires a multidisciplinary approach to improve quality of life and prevent recurrence.

Source

- **Logan and Turner**
- **Scott-brownn**